PADHAKU
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CHAPTER 6

Measuring Space: Perimeter and Area

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ Introduction · The 4 × 100 m relay race

? **In a 4 × 100 m relay, athletes in the outer lanes start ahead of those in the inner lanes, though the finish line is the same. What could be the reason? Does the stagger give anyone an unfair advantage? On what basis can the organisers work out the stagger?**

✔ ANSWER

On the curved parts of the track, an outer lane is a bigger semicircle, so a runner there would run a longer distance.

The stagger (a start ahead) makes every runner cover exactly the same distance.

So the stagger gives **no unfair advantage**; it removes an unfair disadvantage.

The organisers work it out from the **lengths of the semicircular curves** of each lane, i.e., from the circumference of circles (radius of each lane).

Think and Reflect **My school has a 200 m track instead of a 400 m track. Do we need a smaller stagger (than in the Olympics) for the same 4 × 100 m relay?**

✔ ANSWER

On one semicircular bend, the outer lane is longer by π × (difference in radius) = π × lane width. This does **not** depend on the size of the bend.

So the extra length per bend is the same on both tracks.

But a 400 m race covers 2 bends on a 400 m track and **4 bends** on a 200 m track (two laps).

So **no — the stagger is not smaller; in fact a bigger (about double) stagger is needed** on the 200 m track.

🤔 Think and Reflect · 6.1 Perimeter of a Shape

1 **What is the connection between the question 'What is the perimeter of a circle of radius r?' and the one about the 400 m athletics track?**

✔ ANSWER

The curved parts of the track are semicircles.

To find the distance run on each curve, and so the stagger, we must know the **perimeter (circumference) of a circle**.

⭐ 6.1–6.2 Perimeter of a Circle — The C/D Ratio

? **Is the ratio of circumference (C) to diameter (D) the same for circles of all sizes?**

✔ ANSWER

Yes. When a circle is enlarged, C and D grow in the same proportion, so C/D stays the same.

This fixed ratio is called π (about 3.14).

Home Measurement

Wrap a thin thread 20 times round a cotton reel of diameter D; unwrap it and measure its length L. Find L/(20D). Do you get a ratio between 3 and 4? Between 3.1 and 3.2? How could we estimate the ratio by pure geometry?

💡 SAMPLE ANSWER

L is about 20 circumferences, so L/(20D) ≈ C/D.

Example: D = 2.5 cm, L = 157 cm → 157 ÷ 50 = **3.14**.

Yes, the ratio comes **between 3 and 4**, and with careful measuring **between 3.1 and 3.2**.

By geometry: draw polygons inside and outside the circle; the circumference lies between their perimeters (Archimedes' method).

⭐ C/D's Adventurous Journey

Fig. 6.6

A regular hexagon is inscribed in a circle of radius 1. Can you see why this shows that π > 3?

✔ ANSWER

The hexagon is made of 6 equilateral triangles with side = radius = 1.

So the hexagon's perimeter = 6 × 1 = 6.

The circle goes round the outside of the hexagon, so its circumference is longer: $2\pi \times 1 > 6$.

$\therefore 2\pi > 6$, so $\boldsymbol{\pi > 3}$.

Fig. 6.7 **Why does a diagram of an inscribed and a circumscribed hexagon (circle of radius 1) tell us that π is between 3 and 2√3? (Hint: Use the Baudhāyana–Pythagoras Theorem.)**

✔ ANSWER

Inscribed hexagon: perimeter $6 < 2\pi$, so $\pi > 3$.

Circumscribed hexagon with side a: it is made of 6 equilateral triangles whose height is the radius 1.

Height of an equilateral triangle of side a: $a^2 - (a/2)^2 = 1^2 \rightarrow (3/4)a^2 = 1 \rightarrow a = 2/\sqrt{3}$.

Its perimeter = $6a = 12/\sqrt{3} = 4\sqrt{3}$.

The circle lies inside it, so $2\pi < 4\sqrt{3} \rightarrow \pi < 2\sqrt{3} \approx 3.46$.

So $3 < \pi < 2\sqrt{3}$.

🤔 Think and Reflect · A Closer Look at a 400 m Athletics Track

1 **What is the difference in radius between the first and second lanes? Use Fig. 6.11 to find**

the stagger needed by the runner in the second lane. Will an equal stagger be needed between the third and second lanes?

✔ ANSWER

Difference in radius = lane width = **1.22 m**.

For one lap the runner goes round two semicircles = one full circle.

Lane 1 circle: $2\pi \times 36.8$; Lane 2 circle: $2\pi \times (36.8 + 1.22)$.

Extra distance = $2\pi \times 1.22 = 2 \times 3.1416 \times 1.22 \approx$ **7.67 m**.

So the lane-2 runner starts about **7.67 m ahead** (for a one-lap race).

Lane 3 to lane 2: the radius again increases by 1.22 m, so the extra length is again $2\pi \times 1.22 \approx$ 7.67 m. **Yes, the stagger is equal** between every pair of neighbouring lanes.

📝 Exercise Set 6.1

1 **The perimeter of a circle is 44 cm. What is its radius? (Use $\pi \approx 22/7$.)**

✔ ANSWER

$2\pi r = 44$

$2 \times 22/7 \times r = 44$

$r = 44 \times 7 \div 44 = 7$

Radius = 7 cm

2 **Calculate, correct to 3 significant figures, the circumference of a circle with: (i) radius 7 cm (ii) radius 10 cm (iii) radius 12 cm.**

✔ ANSWER

C = 2πr

(i) 2 × 22/7 × 7 = 44 → **44.0 cm**

(ii) 2 × 22/7 × 10 = 440/7 = 62.857... → **62.9 cm**

(iii) 2 × 22/7 × 12 = 528/7 = 75.428... → **75.4 cm**

3 **Calculate the length of the arc of a circle if: (i) the radius is 3.5 cm and the angle at the centre is 60°, and (ii) the radius is 6.3 m and the angle at the centre is 120°.**

✔ ANSWER

Arc length = 2πr × θ/360

(i) 2 × 22/7 × 3.5 × 60/360 = 22 × 1/6 = 11/3 → **≈ 3.67 cm**

(ii) 2 × 22/7 × 6.3 × 120/360 = 39.6 × 1/3 → **13.2 m**

4 **Find the perimeter of a sector (the curved portion and the two straight portions) of a circle of radius 14 cm and sector angle 75°.**

✔ ANSWER

Arc = 2 × 22/7 × 14 × 75/360 = 88 × 75/360 = 55/3 ≈ 18.33 cm

Two radii = 14 + 14 = 28 cm

Perimeter = 18.33 + 28 = **139/3 ≈ 46.33 cm**

5 **Find the perimeters of the shapes in Fig. 6.14 (arcs are quarter, half or three-quarters of a circle):**

(i) 80 m × 60 m rectangle with semicircles on the 60 m sides

(ii) half ring: outer diameter 12 cm, inner diameter 8 cm

(iii) square of side 10 cm with a semicircle on each side

(iv) equilateral triangle of side 12 cm with a semicircle on each side

(v) plus-shaped figure of five 14 cm squares: semicircles at the four ends, quarter circles at the four inner corners

(vi) semicircle on 28 cm with four small semicircles (alternately down and up) on its base

(vii) right triangle with legs 8 cm and 6 cm, semicircles on all three sides

(viii) semicircle on 12 cm with three semicircles on 4 cm parts inside

(ix) semicircle on 20 cm, with semicircles on the two 10 cm halves (one up inside, one down)

💡 **SAMPLE ANSWER**

(i) 2 straight sides = 2 × 80 = 160 m; two semicircles of diameter 60 m = one circle = 22/7 × 60 = 188.57 m

Perimeter = 160 + 188.57 = **≈ 348.57 m**

(ii) Outer semicircle = π × 6 = 18.86 cm; inner semicircle = π × 4 = 12.57 cm; two straight ends = 2 + 2 = 4 cm

Perimeter = 10π + 4 = 220/7 + 4 = **≈ 35.43 cm**

(iii) 4 semicircles of diameter 10 = 2 full circles = 2 × 22/7 × 10 = **≈ 62.86 cm**

(iv) 3 semicircles of diameter 12 = 3 × π × 6 = 18π = 396/7 = **≈ 56.57 cm**

(v) 4 semicircles of radius 7 = 4 × π × 7 = 28π; 4 quarter circles of radius 14 = one circle = 2π × 14 = 28π

Perimeter = 56π = 56 × 22/7 = **176 cm**

(vi) Big semicircle (radius 14) = π × 14 = 44 cm; 4 small semicircles (diameter 7) = 4 × π × 3.5 = 44 cm

Perimeter = **88 cm**

(vii) Hypotenuse = $\sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ cm

Semicircles = π × (10 + 8 + 6)/2 = 12π = 264/7 = **≈ 37.71 cm**

(viii) Big semicircle = π × 6 = 6π; 3 small = 3 × π × 2 = 6π

Perimeter = 12π = **≈ 37.71 cm**

(ix) Big semicircle (radius 10) = 10π; two small (radius 5) = 5π + 5π = 10π

Perimeter = 20π = 440/7 = **≈ 62.86 cm**

6 **The diameter of a car tyre is 56 cm. (i) How far does the car travel when the tyre completes one revolution? (ii) How many revolutions does the tyre make if the car travels 10 km?**

✔ ANSWER

(i) Distance in one revolution = circumference = 22/7 × 56 = **176 cm**

(ii) 10 km = 10 × 1000 × 100 = 10,00,000 cm

Revolutions = 10,00,000 ÷ 176 = 5681.8...

About 5682 revolutions (5681 complete revolutions)

7 **Find the total perimeter of all the petals in each flower.**

(i) Square of side 14 cm; centres of the arcs are the midpoints of the sides (4 petals)

(ii) Regular hexagon of side 42 cm; centres of the arcs are the vertices (6 petals)

✔ ANSWER

(i) Each arc has radius 7 cm (midpoint of a side to a corner or to the centre) and turns through 90°, so it is a quarter circle.

Each petal = 2 quarter circles; 4 petals = 8 quarter circles = 2 full circles

Total = 2 × 2 × 22/7 × 7 = **88 cm**

(ii) Each arc has radius 42 cm (in a regular hexagon, side = distance from vertex to centre) and turns through 60° (equilateral triangle).

One arc = $2 \times 22/7 \times 42 \times 60/360 = 44$ cm

Each petal = 2 arcs = 88 cm; 6 petals = 6×88 = **528 cm**

8 **The ratio of the perimeters of two circles is 5:4. What is the ratio of their radii?**

✔ ANSWER

$2\pi r_1 : 2\pi r_2 = 5 : 4$

Cancel 2π: $r_1 : r_2 =$ **5 : 4**

🤔 Think and Reflect · 6.7 Area of a Parallelogram

1 **What happens if the parallelogram is 'thin' and the foot of the perpendicular from C to AD does not lie on side AD? How do we fix this 'gap'?**

✔ ANSWER

Then cutting off one triangle does not give a rectangle directly.

Fix: take D' on DA and A' on DA extended with A'A = D'D. $\Delta CDD' \cong \Delta BAA'$, so the new parallelogram A'BCD' has the same base, height and area.

Each step makes the parallelogram less slanted. Repeat until the foot of the perpendicular falls on the top side; then the usual cut works.

So **area = base × height** for every parallelogram.

2 **Can we find the area of a parallelogram when we know only the lengths of its sides? Why or why not?**

✔ ANSWER

No. Keep the sides fixed and change the angle between them.

The height changes, so the area (base × height) changes: largest when the angle is 90° (a rectangle), and nearly 0 when the angle is very small.

So we also need an angle or the height.

❓ 6.8 Area of a Triangle · In-text questions

? **What if angle EFG is obtuse (Fig. 6.20B), so the height from E falls outside the base FG? Work out the area.**

✔ ANSWER

Extend GF to the foot I of the perpendicular from E. Let EI = h, FG = b, IF = x.

Area (ΔEIG) = ½ × (x + b) × h

Area (ΔEIF) = ½ × x × h

Area (ΔEFG) = ½(x + b)h − ½xh = **½ bh**

So the formula still works.

? **Why do two congruent triangles ABC and A'B'C' fit together to make a parallelogram (Fig. 6.21)?**

✔ **ANSWER**

Turn one copy through 180° and join it along a common side.

Then equal angles such as ∠B'C'A' and ∠BCA sit in alternate-angle positions.

Equal alternate angles mean the lines are parallel, so both pairs of opposite sides are parallel.

Hence the shape is a **parallelogram** (area bh), and each triangle is half of it: ½ bh.

🤔 Think and Reflect · 6.8 Area of a Triangle

1 **ΔABD and ΔACD (made by median AD) have equal area. Can we cut ΔABD with straight cuts into pieces and rearrange them to exactly cover ΔACD?**

💡 **SAMPLE ANSWER**

Yes, it is possible.

One way: cut each triangle along the line joining the midpoints of its two slanted sides and turn the top piece through 180°; each becomes a parallelogram on the same base (BD = DC) with height h/2.

Two parallelograms with equal base and height can be cut into each other (cut off a triangle at one end and move it to the other end).

Try to find the least number of pieces yourself.

2 **Two polygons P and Q have equal area. Can one always be cut into pieces (straight cuts) that cover the other? Try:**

1. A square and a non-square rectangle with equal area

2. Two triangles with different shapes but equal area

3. A triangle and a square with equal area. Make a conjecture.

💡 **SAMPLE ANSWER**

1. Yes: Baudhāyana's method (6.9) turns a rectangle into a square by cutting.

2. Yes: each triangle can be cut into a rectangle; equal rectangles can be cut into each other.

3. Yes: triangle → rectangle → square.

Conjecture: any two polygons with equal area can be cut into finitely many pieces that rearrange into each other. (This is true; it is called the Wallace–Bolyai–Gerwien theorem.)

3 **Think of rectangles with perimeter 40 units (sides need not be integers).**

1. How many such rectangles are there?

2. Is there one with the largest area? What are its dimensions?

3. Is there one with the smallest area? Are you surprised?

✔ ANSWER

Length + width = 20.

1. **Infinitely many** (length can be any number between 0 and 20, e.g. 15 × 5, 12.5 × 7.5, 19.9 × 0.1).

2. **Yes: the square 10 × 10**, area 100 sq. units. (Area = $l(20 - l) = 100 - (l - 10)^2$, largest when $l = 10$.)

3. **No smallest one.** Making the width tiny (0.1, 0.01, …) makes the area as small as we like, but never 0, so no rectangle has the least area.

❓ Brahmagupta's Formula · In-text question

? **Please check Brahmagupta's formula against other special cases.**

💡 **SAMPLE ANSWER**

Square of side a: s = ½(4a) = 2a

Area = √((2a − a)(2a − a)(2a − a)(2a − a)) = √(a^4) = **a^2** ✓

Rectangle 3 × 4 (example): s = 7, Area = √(4 × 3 × 4 × 3) = √144 = **12** = 3 × 4 ✓

🤔 Think and Reflect · 6.9 Squaring a Rectangle

1 **What procedure would you use to square a given triangle (construct a square with the same area)?**

✔ **ANSWER**

Step 1: Turn the triangle into a rectangle of equal area: base b and height h/2 (cut along the line joining the midpoints of the two other sides and fit the pieces), since ½bh = b × h/2.

Step 2: Square this rectangle with Baudhāyana's construction (Fig. 6.30).

The square has area **½ bh**, the area of the triangle.

📝 Exercise Set 6.2

1 **Find the area of triangle ADE in Fig. 6.31 (rectangle ABCD with AB = DC = 10 cm, BC = 8 cm; E is on BC).**

✔ ANSWER

Take AD = 8 cm as the base.

E lies on BC, which is parallel to AD at a distance DC = 10 cm, so the height = 10 cm.

Area = ½ × 8 × 10 = **40 cm²**

(It is half the rectangle, wherever E is on BC.)

2 **The parallel sides of a trapezium are 40 cm and 20 cm. Its non-parallel sides are equal, each 26 cm. Find the area.**

✔ ANSWER

Drop perpendiculars from the ends of the 20 cm side. Each side piece = (40 − 20) ÷ 2 = 10 cm.

Height h = $\sqrt{(26^2 - 10^2)} = \sqrt{(676 - 100)} = \sqrt{576}$ = 24 cm

Area = ½ × (40 + 20) × 24 = **720 cm²**

3 **Find the area of a triangle whose two sides are 8 cm and 11 cm and perimeter is 32 cm.**

✔ ANSWER

Third side = 32 − 8 − 11 = 13 cm

s = 32 ÷ 2 = 16

Area = √(16 × (16 − 8) × (16 − 11) × (16 − 13)) = √(16 × 8 × 5 × 3) = √1920

= √(64 × 30) = **8√30 ≈ 43.82 cm²**

4 **The sides of a triangular plot are in the ratio 3 : 5 : 7 and its perimeter is 300 m. Find its area.**

✔ ANSWER

3x + 5x + 7x = 300 → x = 20; sides 60 m, 100 m, 140 m

s = 150

Area = √(150 × 90 × 50 × 10) = √6750000 = √(2250000 × 3) = 1500√3

≈ 2598 m² (1500√3 m²)

5 **One diagonal of a rhombus is twice the other. Its area is 128 cm². Find the shorter diagonal.**

✔ ANSWER

Let the diagonals be d and 2d. Area of rhombus = ½ × d_1 × d_2

½ × d × 2d = d^2 = 128

d = √128 = **8√2 ≈ 11.31 cm**

6 **ABCD is a parallelogram. P and Q are any two points on side AB. What can you say about area (ΔPCD) : area (ΔQCD)?**

✔ ANSWER

Both triangles have the same base CD.

P and Q lie on AB, which is parallel to CD, so both have the same height (distance between AB and CD).

So the areas are equal: ratio = **1 : 1**

7 **O is any point on the diagonal PR of a parallelogram PQRS. Prove that the areas of triangles PSO and PQO are equal.**

✔ ANSWER

1. Draw SX ⊥ PR and QY ⊥ PR.

2. Diagonal PR divides the parallelogram into ΔPSR ≅ ΔRQP, so area (ΔPSR) = area (ΔPQR).

3. Both have base PR, so their heights are equal: SX = QY.

4. Area (ΔPSO) = ½ × PO × SX and area (ΔPQO) = ½ × PO × QY.

5. Since SX = QY, **area (ΔPSO) = area (ΔPQO)**.

8 **If the midpoints of the sides of a 4-gon are joined in order, prove that the area of the parallelogram formed is half the area of the 4-gon.**

✔ ANSWER

Let ABCD be the 4-gon and P, Q, R, S the midpoints of AB, BC, CD, DA.

1. In ΔABD, DP is a median, so area (ΔAPD) = ½ area (ΔABD).

2. In ΔAPD, PS is a median, so area (ΔAPS) = ½ area (ΔAPD) = ¼ area (ΔABD).

3. Similarly area (ΔCRQ) = ¼ area (ΔCBD).

4. Adding: area (ΔAPS) + area (ΔCRQ) = ¼ [area (ΔABD) + area (ΔCBD)] = ¼ area (ABCD).

5. In the same way (using diagonal AC): area (ΔBPQ) + area (ΔDRS) = ¼ area (ABCD).

6. The four corner triangles together = ½ area (ABCD).

7. So **area (PQRS) = area (ABCD) − ½ area (ABCD) = ½ area (ABCD)**.

9 **In ΔABC, D is the midpoint of BC and AD is the median. P is any point on AD. Show that area (ΔABP) = area (ΔACP).**

✔ ANSWER

1. AD is a median of ΔABC, so area (ΔABD) = area (ΔACD).

2. PD is a median of ΔPBC, so area (ΔPBD) = area (ΔPCD).

3. Subtract (2) from (1): area (ΔABD) − area (ΔPBD) = area (ΔACD) − area (ΔPCD).

4. So **area (ΔABP) = area (ΔACP)**.

10 **P is a point inside square ABCD. PA, PB, PC, PD are joined. Find the ratio of the area of the red region (ΔPAB and ΔPCD) to the green region (ΔPBC and ΔPDA).**

✔ ANSWER

Let the side be a, and let P be at distances h_1 from AB and h_2 from CD; $h_1 + h_2 = a$.

Red = ½ × a × h_1 + ½ × a × h_2 = ½ a ($h_1 + h_2$) = ½ a^2

In the same way, green = ½ a^2.

Ratio = **1 : 1**

11 **In ΔABC, D is the midpoint of AB. P is any point on BC, and Q is a point on AB such that CQ ∥ PD. PQ is joined. Prove that area (ΔBPQ) = ½ area (ΔABC).**

✔ ANSWER

1. ΔDPQ and ΔDPC have the same base DP, and Q, C lie on the line QC parallel to DP, so they have the same height. Hence area (ΔDPQ) = area (ΔDPC).

2. Area (ΔBPQ) = area (ΔBPD) + area (ΔDPQ)

= area (ΔBPD) + area (ΔDPC) = area (ΔBDC).

3. CD is a median of ΔABC (D is the midpoint of AB), so area (ΔBDC) = ½ area (ΔABC).

4. Hence **area (ΔBPQ) = ½ area (ΔABC)**.

🤔 Think and Reflect · 6.10 Area of a Circle

1 **Why were human beings so fond of circular shapes? Only for practical reasons, or other reasons too? What uses have they found for the circle?**

💡 **SAMPLE ANSWER**

Practical: a wheel rolls smoothly; a circle encloses the most area for a given boundary length, so it saves material; circular walls and huts are strong; easy to draw with a rope and a peg.

Other reasons: beauty and symmetry, religious and cultural meanings (sun, moon, cycles of life).

Uses: wheels, pots, wells, towers, coins, rangoli, clocks, gears, round houses and stadiums.

? **When the circle's slices are rearranged (Fig. 6.37), why is the base of the parallelogram half the circumference?**

✔ **ANSWER**

The arcs of the slices together make the whole circumference $2\pi r$.

Half of the slices point up and half point down, so the top edge and the bottom edge each get half of the arcs.

So base = ½ × $2\pi r$ = **πr**, height = r, and area = $\pi r \times r = \pi r^2$.

📝 Exercise Set 6.3

1 **Find the area of a sector of a circle with radius 7 cm and sector angle 60°.**

✔ ANSWER

Area = $\pi r^2 \times \theta/360$ = $22/7 \times 7 \times 7 \times 60/360$

= $154 \times 1/6 = 77/3$ ≈ **25.67 cm²**

2 **Find the area of a quadrant of a circle whose circumference is 44 cm.**

✔ ANSWER

$2\pi r = 44 \rightarrow r = 7$ cm

Quadrant = ¼ × $22/7 \times 7 \times 7 = 154 \div 4$ = **38.5 cm²**

3 **The minute hand of a clock is 7 cm long. Find the area swept by it in 10 minutes.**

✔ ANSWER

In 60 minutes it turns 360°, so in 10 minutes it turns 60°.

Area = 22/7 × 7 × 7 × 60/360 = 77/3 ≈ **25.67 cm²**

4 **A chord of a circle of radius 10 cm subtends 90° at the centre. Find the area of the (i) minor sector (90°), (ii) major sector (270°). (Use π ≈ 3.14.)**

✔ ANSWER

Circle = 3.14 × 10 × 10 = 314 cm²

(i) Minor sector = 314 × 90/360 = **78.5 cm²**

(ii) Major sector = 314 × 270/360 = **235.5 cm²**

5 **A chord of a circle of radius 15 cm subtends 60° at the centre. Find the areas of the minor and major segments. (Use π ≈ 3.14 and √3 ≈ 1.73.)**

✔ ANSWER

Sector = 3.14 × 15 × 15 × 60/360 = 706.5 ÷ 6 = 117.75 cm²

The triangle formed by the two radii and the chord is equilateral (60°, two sides 15 cm).

Triangle = (√3/4) × 15² = 1.73 × 225 ÷ 4 = 97.3125 cm²

Minor segment = 117.75 − 97.3125 = **≈ 20.44 cm²**

Circle = 3.14 × 225 = 706.5 cm²

Major segment = 706.5 − 20.4375 = **≈ 686.06 cm²**

6 **A car has two wipers which do not overlap. Each blade is 28 cm long and sweeps 120°. Find the total area cleaned at each sweep.**

✔ ANSWER

One wiper = 22/7 × 28 × 28 × 120/360 = 2464 × 1/3 = 821.33 cm²

Two wipers = 2 × 2464/3 = 4928/3 **≈ 1642.67 cm²**

***7** **A chord of a circle of radius r subtends 60° at the centre. Show that the area of the minor segment is $r^2(\pi/6 - \sqrt{3}/4)$.**

✔ ANSWER

Sector = $\pi r^2 \times 60/360 = \pi r^2/6$

The two radii and the chord make an equilateral triangle of side r; its area = $(\sqrt{3}/4)\, r^2$.

Minor segment = sector − triangle = $\pi r^2/6 - (\sqrt{3}/4)\, r^2$

= **$r^2\,(\pi/6 - \sqrt{3}/4)$** = $\pi r^2\,(1/6 - \sqrt{3}/(4\pi))$

(Note: the book prints $\pi r^2(1/6 - \sqrt{3}/4)$; the π should not multiply the √3/4 term.)

***8** **An equilateral triangle is inscribed in a circle of radius r. Show that area of triangle : area of circle = 3√3/(4π) ≈ 0.413.**

✔ ANSWER

Join the centre O to the vertices: 3 triangles, each with two sides r and angle 120° at O.

Side of the triangle: drop a perpendicular from O to a side; it makes 30°-60°-90° triangles, half-side = r × √3/2, so side = √3 r.

Area of triangle = (√3/4)(√3 r)² = (√3/4) × 3r² = 3√3 r²/4

Ratio = (3√3 r²/4) ÷ πr² = **3√3/(4π) ≈ 5.196 ÷ 12.566 ≈ 0.413**

***9** **A square is inscribed in a circle of radius r. Show that area of square : area of circle = 2/π ≈ 0.637.**

✔ ANSWER

The diagonal of the square is a diameter = 2r.

Area of square = ½ × d × d = ½ × 2r × 2r = 2r²

Ratio = 2r² ÷ πr² = **2/π ≈ 0.637**

***10** **A (regular) hexagon is inscribed in a circle of radius r. Show that area of hexagon : area of circle = 3√3/(2π) ≈ 0.827. Why is this exactly twice the answer to Question 8?**

✔ ANSWER

The hexagon is made of 6 equilateral triangles of side r.

Area = $6 \times (\sqrt{3}/4)\, r^2 = 3\sqrt{3}\, r^2/2$

Ratio = $(3\sqrt{3}\, r^2/2) \div \pi r^2 = \mathbf{3\sqrt{3}/(2\pi) \approx 0.827}$

Why twice: join alternate vertices A, C, E of the hexagon to get the inscribed equilateral triangle. It is made of 3 triangles OAC, OCE, OEA, each with area $\frac{1}{2} r^2 \sin 120° = (\sqrt{3}/4)\, r^2$. So triangle = $3 \times (\sqrt{3}/4)\, r^2$, while hexagon = $6 \times (\sqrt{3}/4)\, r^2$, exactly double.

End-of-Chapter Exercises

1 **The area figure for $(a + b)^2 = a^2 + 2ab + b^2$ is a square of side a + b cut into a^2, ab, ab, b^2. Draw figures for $(a + b)(a - b) = a^2 - b^2$ and $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.**

✔ ANSWER

$(a + b)(a - b) = a^2 - b^2$:

- Draw a square of side a and cut away a square of side b from one corner. The L-shape left has area $a^2 - b^2$.
- Cut the L-shape into two rectangles: $a \times (a - b)$ and $b \times (a - b)$.
- Place them end to end: one rectangle of length a + b and width a − b. So $(a + b)(a - b) = a^2 - b^2$.

$(a + b + c)^2$:

- Draw a square of side a + b + c; divide each side into parts a, b, c and draw lines across.

- You get 9 pieces: squares a^2, b^2, c^2 on the diagonal and two each of rectangles ab, bc, ca.
- So $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.

2 **An isosceles triangle has perimeter 40 cm; the equal sides are 15 cm each. Find its area.**

✔ ANSWER

Base = 40 − 15 − 15 = 10 cm

Height = $\sqrt{(15^2 - 5^2)} = \sqrt{(225 - 25)} = \sqrt{200} = 10\sqrt{2}$ cm

Area = ½ × 10 × $10\sqrt{2}$ = **$50\sqrt{2} \approx 70.71$ cm²**

(Heron check: s = 20, $\sqrt{(20 \times 5 \times 5 \times 10)} = \sqrt{5000} = 50\sqrt{2}$ ✓)

3 **An isosceles triangle has base 10 cm and area 60 cm². What are the lengths of the equal sides?**

✔ ANSWER

½ × 10 × h = 60 → h = 12 cm

Half the base = 5 cm

Equal side = $\sqrt{(5^2 + 12^2)} = \sqrt{169}$ = **13 cm**

4 **The area of a right-angled triangle is 54 sq. cm. One leg is 12 cm. Find its perimeter.**

✔ ANSWER

½ × 12 × other leg = 54 → other leg = 9 cm

Hypotenuse = $\sqrt{(12^2 + 9^2)} = \sqrt{225} = 15$ cm

Perimeter = 12 + 9 + 15 = **36 cm**

5 **The sides of a triangle are in the ratio 2 : 3 : 4 and its perimeter is 45 cm. Find its area.**

✔ ANSWER

9x = 45 → x = 5; sides 10, 15, 20 cm

s = 22.5; s − a = 12.5, s − b = 7.5, s − c = 2.5

Area = $\sqrt{(22.5 \times 12.5 \times 7.5 \times 2.5)} = \sqrt{(84375/16)} = 75\sqrt{15}/4$

≈ 72.62 cm²

6 **The sides of a triangle are 7 cm, 24 cm, 25 cm. Find its area in two different ways.**

✔ ANSWER

Way 1: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$, so the triangle is right-angled (legs 7 and 24).

Area = ½ × 7 × 24 = **84 cm²**

Way 2 (Heron): s = (7 + 24 + 25)/2 = 28

Area = $\sqrt{(28 \times 21 \times 4 \times 3)} = \sqrt{7056}$ = **84 cm²**

7 **A bicycle wheel has diameter 60 cm. How far does the cyclist travel when the wheel rotates 100 times?**

✔ ANSWER

One rotation = πd = 22/7 × 60 = 1320/7 ≈ 188.57 cm

100 rotations = 132000/7 ≈ 18857 cm

≈ 188.57 m

8 **Find the area of a quadrant of a circle whose circumference is 66 cm.**

✔ ANSWER

2 × 22/7 × r = 66 → r = 66 × 7 ÷ 44 = 10.5 cm

Quadrant = ¼ × 22/7 × 10.5 × 10.5 = 346.5 ÷ 4

≈ 86.63 cm² (86.625 cm²)

9 **The wheel of a car has an outer radius of 28 cm. How far does the car travel in one complete turn, and how many turns does the wheel make in 1 km?**

✔ ANSWER

One turn = 2 × 22/7 × 28 = **176 cm**

1 km = 1,00,000 cm

Turns = 1,00,000 ÷ 176 = 568.18...

About 568 turns

***10** **Two rectangles have the same area and the same perimeter. Are they congruent?**

✔ ANSWER

Let the sides be l_1, w_1 and l_2, w_2 with $l_1 + w_1 = l_2 + w_2 = S$ and $l_1w_1 = l_2w_2 = A$.

$(l_1 - w_1)^2 = (l_1 + w_1)^2 - 4l_1w_1 = S^2 - 4A$, and the same for $l_2 - w_2$.

So $l_1 - w_1 = l_2 - w_2$ (taking the longer side first).

Same sum and same difference → $l_1 = l_2$ and $w_1 = w_2$.

Yes, they are congruent.

11 **Using area of a parallelogram = base × height and Fig. 6.42 (trapezium with parallel sides a and b, height h, cut into a parallelogram and a triangle), show that area of a trapezium = ½ (a + b)h.**

✔ ANSWER

Draw a line from one end of the top side parallel to the slanting side.

It cuts the trapezium into a parallelogram (base a, height h) and a triangle (base b − a, height h).

Area = ah + ½ (b − a)h = ½ (2a + b − a)h = **½ (a + b)h**

12 **By dividing a trapezium into two triangles, show that its area is ½ (a + b)h.**

✔ ANSWER

Draw a diagonal. It makes one triangle with base a and one with base b.

Both have height h (the distance between the parallel sides).

Area = ½ ah + ½ bh = **½ (a + b)h**

13 **Show how two identical copies of a trapezium make a parallelogram. How does this give the area formula?**

✔ ANSWER

Turn the second copy through 180° and join it to the first along a slanting side.

The top becomes a + b and the bottom b + a; the slanting sides are parallel, so we get a parallelogram with base (a + b) and height h.

Its area = (a + b)h, and it is two trapeziums.

So one trapezium = **½ (a + b)h**

14 **Show that the area of a kite is half the product of its diagonals: (i) using algebra, (ii) using geometry.**

✔ ANSWER

Kite ABCD with AB = AD, CB = CD; diagonal AC (= d_1) is the perpendicular bisector of diagonal BD (= d_2); they meet at O.

(i) Algebra: Area = area (ΔABC) + area (ΔADC)

= ½ × d_1 × BO + ½ × d_1 × OD

= ½ d_1 (BO + OD) = **½ d_1 d_2**

(ii) Geometry: through each vertex draw a line parallel to the other diagonal. This makes a rectangle d_1 × d_2 around the kite.

The diagonals cut the rectangle into 4 small rectangles; the kite takes exactly half of each (a diagonal halves a rectangle).

So kite = ½ × rectangle = ½ $d_1 d_2$

15 (i) **Rectangle ABCD has sides a, b and rectangle PQRS has sides 2a, 2b. Show PQRS has 4 times the area. Do 4 copies of ABCD fit into PQRS?**

✔ ANSWER

Area (PQRS) = 2a × 2b = 4ab = 4 × area (ABCD).

Yes: halve each side of PQRS and join; we get 4 rectangles a × b in a 2 × 2 pattern.

15 (ii) **ΔABC has sides a, b, c and ΔPQR has sides 2a, 2b, 2c. Show ΔPQR has 4 times the area. Do 4 copies of ΔABC fit into ΔPQR?**

✔ ANSWER

Semi-perimeter of ΔPQR = 2s, and each bracket doubles: 2s − 2a = 2(s − a), etc.

Area (ΔPQR) = √(2s × 2(s − a) × 2(s − b) × 2(s − c)) = √16 × √(s(s − a)(s − b)(s − c)) = 4 × area (ΔABC).

Yes: join the midpoints of the sides of ΔPQR. We get 4 triangles each with sides a, b, c, so each is congruent to ΔABC (SSS).

15 (iii) **ΔABC has sides a, b, c and ΔPQR has sides 3a, 3b, 3c. Show ΔPQR has 9 times the area. Do 9 copies of ΔABC fit into ΔPQR?**

✔ ANSWER

Semi-perimeter = 3s, each bracket triples.

Area (ΔPQR) = √(81 × s(s − a)(s − b)(s − c)) = 9 × area (ΔABC).

Yes: divide each side into 3 equal parts and draw lines through these points parallel to the sides. We get 9 small triangles (6 one way up, 3 upside down), each with sides a, b, c.

***16** **Fig. 6.43: In a triangle, one side is bisected and another side is trisected; the shaded 4-gon has vertices: the common vertex at the base, the midpoint, and the two trisection points. What fraction of the triangle is shaded?**

Fig. 6.44: In a square, each vertex is joined to the midpoint of a side; the lines form a small square in the middle. What fraction of the square is shaded?

💡 SAMPLE ANSWER

Fig. 6.43: Let the top vertex be T, the bottom-left vertex L, the bottom-right vertex R; P = midpoint of TL; Q_1, Q_2 trisect TR ($TQ_1 = Q_1Q_2 = Q_2R$). Let area (ΔTLR) = S.

Area (ΔTLQ_2) = (TQ_2/TR) × S = ⅔ S (same height from L).

Area (ΔTPQ_1) = ½ × ⅓ × S = ⅙ S.

Shaded = ⅔ S − ⅙ S = ½ S → **½ of the triangle**

Fig. 6.44: Take the square with side 2 and corners (0, 0), (2, 0), (2, 2), (0, 2). One line joins (0, 0) to (2, 1): $x - 2y = 0$; the parallel line joins (0, 1) to (2, 2): $x - 2y + 2 = 0$.

Distance between them = $2/\sqrt{5}$ = side of the small square.

Small square = 4/5; big square = 4. Fraction = **1/5**

17 **Fig. 6.45: 3 equal circles fit in a row in a rectangle. Fig. 6.46: 4 equal circles fit in a row in a rectangle. What fraction of each rectangle is covered by the circles?**

✔ ANSWER

Let the radius be r. The rectangle's height is 2r.

3 circles: rectangle $6r \times 2r = 12r^2$; circles $3\pi r^2$.
Fraction = $3\pi r^2/12r^2$ = **π/4 ≈ 0.785**

4 circles: rectangle $8r \times 2r = 16r^2$; circles $4\pi r^2$.
Fraction = **π/4** again (≈ 11/14 using 22/7).

18 **Make a conjecture about the area occupied by circles fitted into a rectangle in this way. Test it for 10, 20, 50 circles, then prove it.**

✔ ANSWER

Conjecture: however many circles there are, they cover **π/4** of the rectangle (about 78.5 %).

10 circles: $10\pi r^2 \div (20r \times 2r) = 10\pi r^2/40r^2 = \pi/4$

20 circles: $20\pi r^2/80r^2 = \pi/4$; 50 circles: $50\pi r^2/200r^2 = \pi/4$

Proof: n circles of radius r need a rectangle $2nr \times 2r = 4nr^2$. Circles' area = $n\pi r^2$. Fraction = $n\pi r^2/4nr^2 = \pi/4$, which does not depend on n.

***19** **Nine identical rectangles fit together to make a large rectangle of area 72 cm² (Fig. 6.47: 4 lying flat on top, 5 standing below). Find the perimeter of each small rectangle.**

💡 **SAMPLE ANSWER**

Let each small rectangle be L (long) × w (short).

Top row: 4 lengths; bottom row: 5 widths. So $4L = 5w \rightarrow w = 4L/5$.

Area: $9 \times L \times w = 72 \rightarrow Lw = 8$

$L \times 4L/5 = 8 \rightarrow L^2 = 10 \rightarrow L = \sqrt{10} \approx 3.16$ cm; $w = 4\sqrt{10}/5 \approx 2.53$ cm

Perimeter $= 2(L + w) = 2 \times 9\sqrt{10}/5 = 18\sqrt{10}/5$

$\approx$ 11.38 cm

***20** **A vertex of a triangle is joined to the points of trisection of the opposite side (Fig. 6.48). Show that the shaded blue triangle (first part) and red triangle (third part) have equal areas. Find a way to cut the blue triangle into pieces that cover the red triangle.**

💡 **SAMPLE ANSWER**

Both triangles have equal bases (each ⅓ of the side) on the same line, and the same height (from the common vertex).

So their areas (½ × base × height) are **equal**.

Cutting (one way): cut the blue triangle along the line joining the midpoints of its two slanting sides and turn the top piece through 180° → a parallelogram with the same base and half the height.

The red triangle can be made into a parallelogram with the same base and height in the same way; parallelograms with equal base and height can be cut into each other (cut off a triangle at one end, move it to the other end).

Reverse the red steps to finish the red triangle.

***21** **A quarter circle in a square has its centre at one vertex and passes through two adjacent vertices. Two semicircles are drawn on two adjacent sides as diameters, creating shaded regions A (where the semicircles overlap) and B (inside the quarter circle,**

outside both semicircles). Show that A and B have equal area.

✔ ANSWER

Let the side of the square be 2a.

Quarter circle = ¼ × π(2a)² = πa²

Each semicircle (radius a) = ½ πa²; two semicircles = πa²

The region covered by the two semicircles = πa² − A (A is counted twice).

B = quarter circle − covered region = πa² − (πa² − A) = A

So area A = area B.

*22 **Four semicircles are drawn inside a square of side 2 units, with centres at the midpoints of the sides. They form a 4-petalled flower. Find its perimeter and area.**

✔ ANSWER

Each semicircle has radius 1.

Each petal is bounded by two quarter circles of radius 1 (from a corner of the square to its centre).

Perimeter = 8 quarter circles = 2 full circles = 2 × 2π × 1 = **4π ≈ 12.57 units**

Area: the 4 semicircles together = 4 × ½ π × 1² = 2π. They cover the square (4) and cover each petal twice.

So $2\pi = 4 +$ petals $\rightarrow$ petals = **$2\pi - 4 \approx 2.28$ sq. units**

***23** **Two concentric circles have centre O. Chord BC of the larger circle touches the smaller circle at A, and BC = l. Show that the area between the circles is ¼ πl^2.**

✔ ANSWER

Let the radii be R (larger) and r (smaller).

OA $\perp$ BC (radius to a tangent), and the perpendicular from the centre bisects the chord: BA = l/2.

In right ΔOAB: $R^2 = r^2 + (l/2)^2 \rightarrow R^2 - r^2 = l^2/4$

Area of ring = $\pi R^2 - \pi r^2 = \pi(R^2 - r^2) =$ **¼ πl^2**

***24** **Semicircles are drawn on all the sides of a right-angled triangle (Fig. 6.52). Show that Area (A) + Area (B) = Area (C).**

✔ ANSWER

Let the legs be a, b and hypotenuse c, with $a^2 + b^2 = c^2$. C is the triangle; A and B are the crescents outside the semicircle on the hypotenuse.

Semicircles: on a: $\pi a^2/8$; on b: $\pi b^2/8$; on c: $\pi c^2/8$.

So semicircle(a) + semicircle(b) = $\pi(a^2 + b^2)/8 = \pi c^2/8$ = semicircle(c).

The whole figure = triangle + semicircle(a) + semicircle(b) = semicircle(c) + A + B.

Since semicircle(a) + semicircle(b) = semicircle(c): **Area (C) = Area (A) + Area (B)**.

***25** **Two circles of radius r pass through each other's centres (Fig. 6.53). Find the area of the region common to both circles.**

✔ ANSWER

Centres A, B; the circles meet at C and D. ΔACB and ΔADB are equilateral, so ∠CAD = ∠CBD = 120°.

The common region is made of two equal segments cut off by chord CD.

One segment = sector (120°) − ΔACD = $\pi r^2/3 - \frac{1}{2} r^2 \sin 120° = \pi r^2/3 - (\sqrt{3}/4) r^2$

Common region = 2 × segment = **$(2\pi/3 - \sqrt{3}/2) r^2 \approx 1.23\, r^2$**

***26** **Three triangles of areas A, B, C lie in a rectangle (Fig. 6.54). Show that the area of the rectangle is 2(A + C)(B + C)/C.**

💡 SAMPLE ANSWER

Let the rectangle be W × H with the bottom-left corner O. The top point is Q (distance p from the left side), the right point is R (height q), and the inner point S is directly below Q and level with R.

C (right triangle QSR) = ½ (H − q)(W − p)

A (triangle OQS, base QS = H − q, height p) = ½ p (H − q)

B (triangle OSR, base SR = W − p, height q) = ½ q (W − p)

A + C = ½ (H − q) W and B + C = ½ (W − p) H

(A + C)(B + C) = ¼ WH (H − q)(W − p) = ¼ WH × 2C

So 2(A + C)(B + C)/C = **WH = area of the rectangle**.

***27** **Fig. 6.55: semicircle on AC with centre O and radius R; B is the top point (OB ⊥ AC). Another semicircle is drawn on AB (centre D). Show that the two shaded regions — the crescent between the two arcs AB, and triangle AOB — have equal areas.**

✔ ANSWER

AB = √(R² + R²) = R√2, so the radius of the small semicircle = R√2/2.

Small semicircle = ½ π (R√2/2)² = πR²/4

Segment between chord AB and the big arc = quarter circle − ΔAOB = πR²/4 − R²/2

Crescent = small semicircle − segment = πR²/4 − (πR²/4 − R²/2) = R²/2

ΔAOB = ½ × R × R = R²/2

So the two shaded regions have equal area.

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**